

ThawHome (Tube)



User Manual

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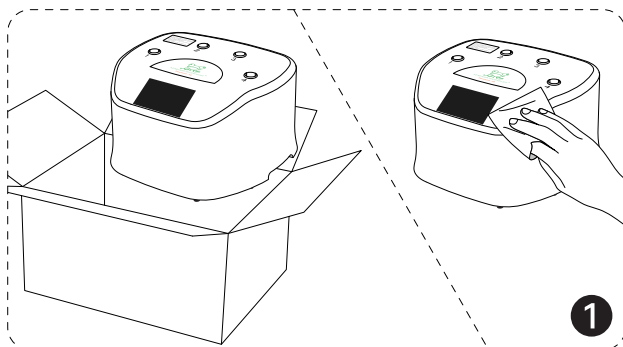
Function description and appearance

In biological laboratories, researchers attach increasing importance to the cryopreservation, resuscitation and freeze of the sample. As the essential steps in the lab, their stability and the polluted rate will have a direct influence on the result of experiments, and thus a set of products will better help researchers conduct experiments and get the research results more quickly and efficiently. Meanwhile, industrial users also need similar products to ensure stable production and increase efficiency.KMS has been focusing on such kind of research and development for ten years, and the products the company has introduced enable the users to better cryopreserve, resuscitate and freeze their samples, providing a better and stabler operation environment.

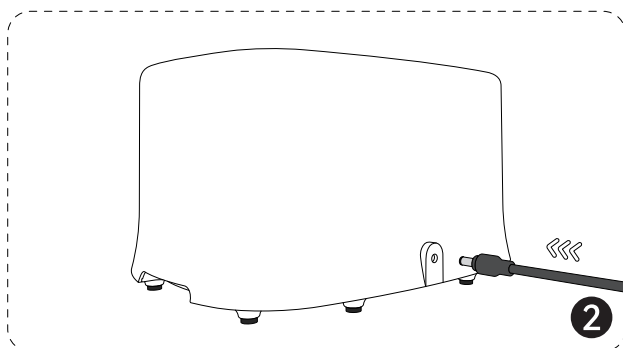
ThawHome is aimed at thawing the active biological cells quickly in the tubes of 1.5ml~2.0ml and 4.0ml~5.0ml.At first,biological tubes that are directly taken out from liquid nitrogen or dry ice need to be insert into the notch. And then, the unfreezing system quickly restore biological vitality through the sensor and temperature control technology, by which the users can resuscitate the frozen cells.



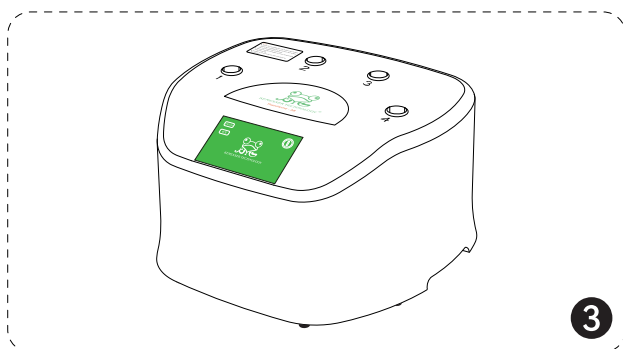
Quick Start Guide



Take the instrument out of the carton and wipe its surface.



Carefully connect the power cord from the accessories to the rear power socket to start the instrument.



Enter the initial interface.



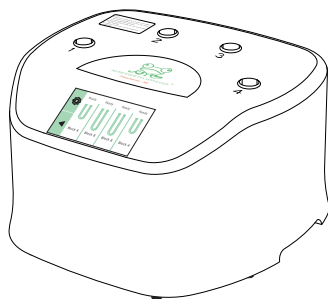
Click the power button at the top right corner of Step ③ interface to enter the preheating interface. The preheating time is about 6~10 minutes.

4



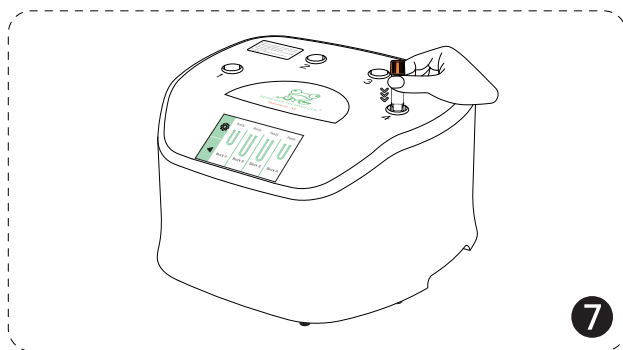
Preheating Completed.

5



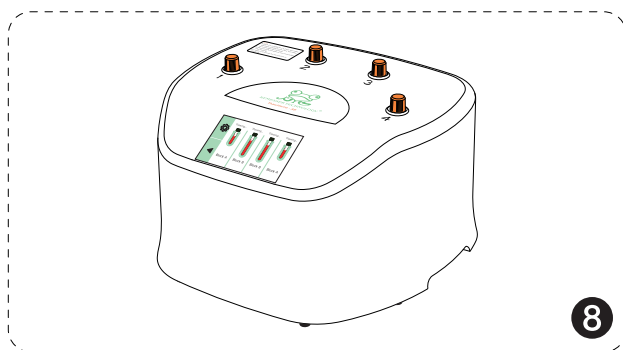
After preheating is completed, enter the main interface, which displays "Ready".

6



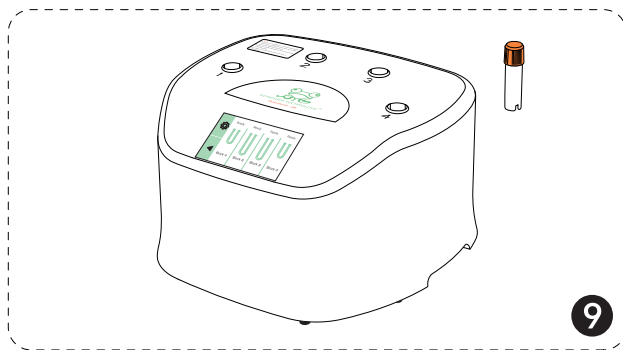
Carefully insert the tube to be thawed into the corresponding thawing slot.

7



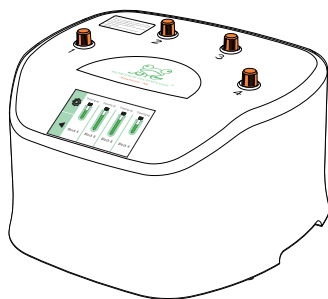
The interface displays "Thawing".

8



After thawing is completed, immediately remove the tube and shake it for a few seconds until no ice is visible in the container, then store it immediately.

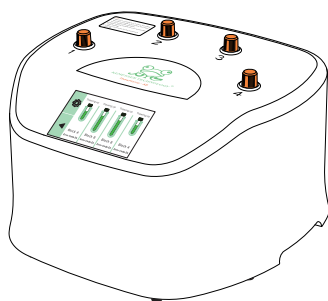
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Beep sound
ten times

10

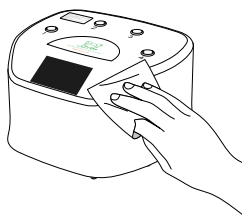
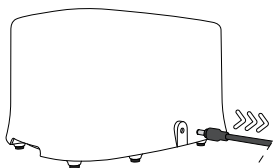
If the sample is not removed within 5 seconds after thawing is completed, the device will sound an alarm with ten beeps.



3-second
long beep

11

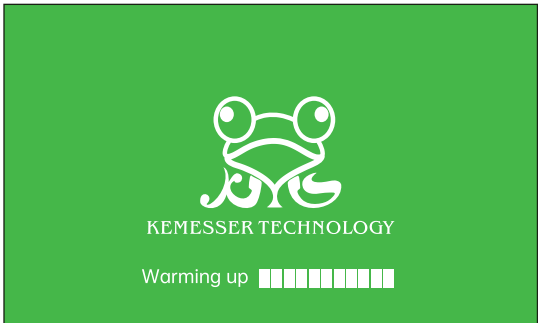
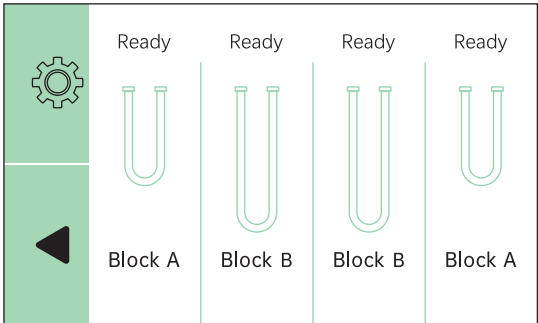
If the sample is not removed within 15 seconds, the device will emit a 3-second long beep alarm and display Error Code E6.

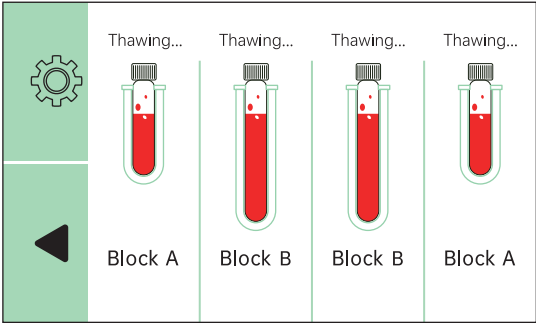
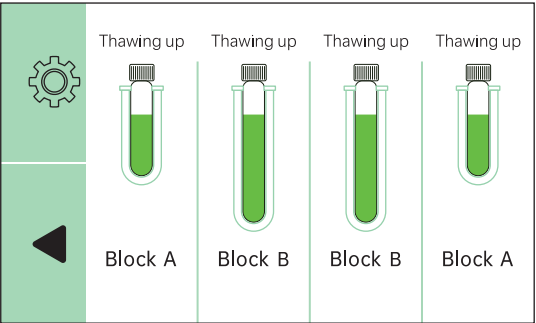
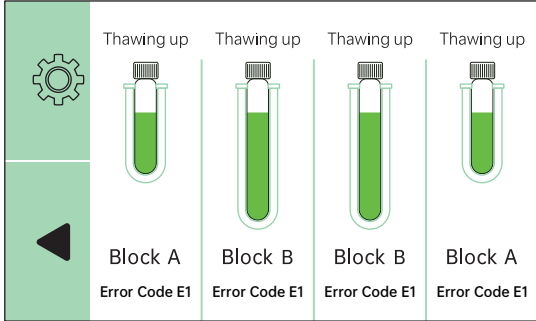


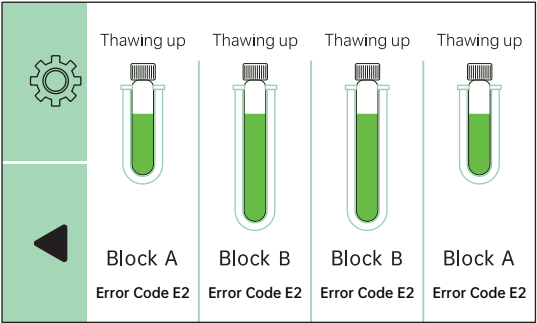
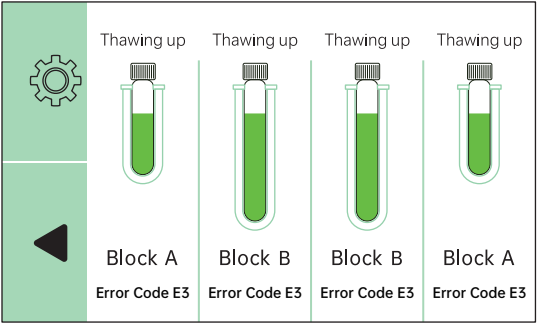
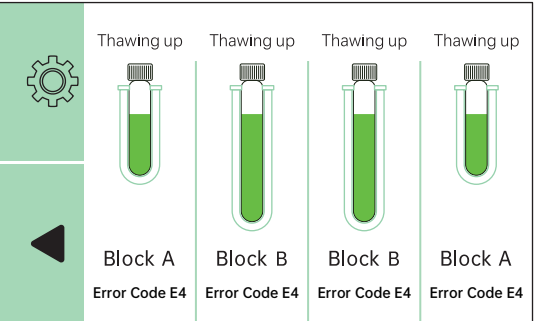
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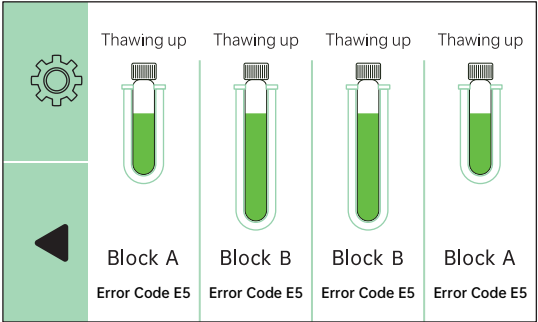
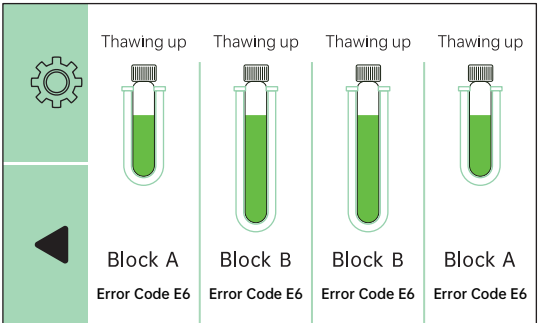
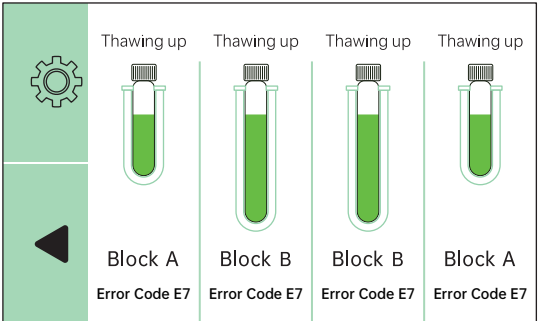
After the experiment were completed, carefully unplug the power cord and wipe the instrument surface before storing the instrument properly.

Software Interface

No.	Steps	Display Screen Indication	Sound
1	Check that the instrument is in good condition and connected to the power supply.	N/A	N/A
2	Initial Interface The power supply is normal, the screen automatically lights up, and then click on the upper left corner to select the Chinese or English interface.		N/A
3	Preheating Interface Click on the upper right power button to enter the preheating, preheating time for about 6~10 minutes.		N/A
4	Main Interface When the preheating is complete, you will enter the operation interface.		N/A

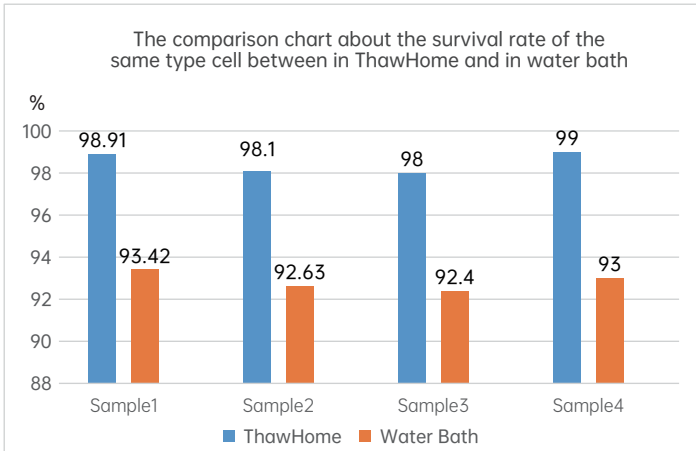
No.	Steps	Display Screen Indication	Sound
5	Working Interface Place the tube sample (below -10 °C) into the corresponding resuscitation slot carefully. Tips: Before freezing, close the lid of the cryogenic bottle after being filled to avoid leakage, clean the bottle mouth with 70% alcohol or spray it on the bottle mouth.	 The display screen shows four test tubes arranged horizontally. Each tube is labeled 'Thawing...' at the top. Below each tube is a label: 'Block A', 'Block B', 'Block B', and 'Block A'. The tubes contain a red liquid. To the left of the tubes is a green vertical bar containing a gear icon (top) and a black triangle icon (bottom).	N/A
6	Working Interface After the resuscitation is complete, the buzzer of the device beeps once, the inside of the resuscitation slot shows green cell fluid, and the interface shows "Thawing up".	 The display screen shows four test tubes arranged horizontally. Each tube is labeled 'Thawing up' at the top. Below each tube is a label: 'Block A', 'Block B', 'Block B', and 'Block A'. The tubes contain a green liquid. To the left of the tubes is a green vertical bar containing a gear icon (top) and a black triangle icon (bottom).	N/A
7	Working Interface If the initial temperature of the sample tube taken from an ultra-cold refrigerator or a liquid nitrogen tank is too high, an alert E1 will be triggered. Remove the overheated tube and insert the cryogenic tube to retry. If the alarm remains to continue, please contact Kemesser for relevant technical support.	 The display screen shows four test tubes arranged horizontally. Each tube is labeled 'Thawing up' at the top. Below each tube is a label: 'Block A', 'Block B', 'Block B', and 'Block A'. The tubes contain a green liquid. Below the 'Block B' labels, the text 'Error Code E1' is displayed. To the left of the tubes is a green vertical bar containing a gear icon (top) and a black triangle icon (bottom).	N/A

No.	Steps	Display Screen Indication	Sound
8	Working Interface The infrared sensor is abnormal, an alert E2 will be triggered. If the infrared sensors is abnormal, please enter the factory settings; if it still reports an error after re-opening, please contact Kemesser for relevant technical support.		N/A
9	Working Interface The temperature of NTC is detected to be abnormal, an alert E3 will be triggered. The temperature of NTC is detected to be abnormal (The temperature is detected as zero or too high or too low), please contact Kemesser for relevant technical support.		N/A
10	Working Interface The heating object works abnormally, an alert E4 will be triggered. The current of the heating object is too high or is zero, and the equipment reports an error E4, please contact Kemesser for relevant technical support and replace the heating device.		N/A

No.	Steps	Display Screen Indication	Sound
11	Working Interface The electromagnet works abnormally, an alert E5 will be triggered. The current of electromagnet is too high or zero to grip the heating device, and the equipment reports an error E5, please contact Kemesser for relevant technical support.		N/A
12	Working Interface The resuscitated samples should be thrown away if the tubes are not taken away after being thawed for 15S, an alert E6 will be triggered. Remove the tube, and the survival rate of the sample can not be guaranteed;		N/A
13	Working Interface The resuscitation process has been complete, but the tube is not ejected or raised, an alert E7 will be triggered. Grab the tube cover gently with your fingers or tweezers and remove it from the opening of the device. Make sure the inner heat pad is intact. Note: If the heat pad is incomplete or damaged, please contact Kemesser for relevant technical support.		N/A

Package Contents and Technical Specifications

Lists	Numbers	Remarks
Cell Thawing Device	1	Equipment
DC Switching Power Supply	1	12V/15A
Operation Introductions	1	
Instrument Parameter		
The unfreezing time of notch A	2~3 minutes	
The unfreezing time of notch B	3~5 minutes	
The temperature of operating environment	5°C~35°C; the room humidity is 75% (max.)	
The storage temperature	-20°C~60°C	
The maximum height	2KM	
Rated power	120W Max.	
The input and output voltage (power adapter)	Alternating current 100-240V /DC12V/50-60HZ	
Output current	15A	
Attaching plug	US,UK,EU	
Size	L x W x H : 296mm x 270mm x 170mm	
Certification	CE	



Safety Instructions

The automatic cell thawing system is designed for trained and experienced laboratory personnel.

1. Please read the instructions before installing, using or cleaning the equipment.
2. To prevent serious or fatal electric shock, please use only the AC power supply that comes with this package.
3. Do not place near water or other liquids, or soak in water or other liquids.
4. Taking the enclosure apart unauthorizedly will invalidate your valid warranty. Do not remove the enclosure unless authorized by Kemesser.
5. There is a DC jack in the back of this machine. Do not insert any DC plug into this port or attempt to connect the device to other devices (such as DC power supply, computer) unless authorized by Kemesser.
6. The equipment is designed for use in indoor bench-top laboratories, so please ensure it to be placed safely in the horizontal when used, and it is not recommended to be used close a centrifuge, an exciter or a vortex mixer when they have a strong vibration.
7. Make sure the tube is frozen in an upright and vertical position. The frozen contents are at the bottom of the tube.
8. Make sure the cryogenic test tube has a clean appearance, and seal the cap tightly to prevent leakage.
9. Insert the cryogenic tube into the opening of the device and start the thawing process. Do not touch the tube with your hand or any other object until the resuscitation process is complete.
10. The time for tube resuscitation in tank A is generally 2~3 minutes, and the time for tube defrosting in Tank B is generally 3~5 minutes. When using, laboratory personnel are recommended standing close to the instrument so that the tubes can be transferred immediately after the completion of resuscitation.
11. Keep the openings of the instrument covered when not in use.
12. When the device is opened, the heating element in the bottle mouth is heated. Avoid inserting body parts (such as fingers or anything other than a cryogenic vial) into the resuscitation notch at all time.

Care and Cleaning

Do not attempt to clean internal components. Do not pour any cleaning objects or solutions into the recovery tank at the top opening of the instrument. Before or after each use, wipe the outside with a soft, non-abrasive laboratory cloth moistened with 70% alcohol. Resuscitating condensate is not a problem because it evaporates during thawing or standby.

Technical Service and Repair

Contact KEMESSER TECHNOLOGY support at 0571-87156759 or email andy@kemesser.com for help.

Certificate of Conformity:

Certificate of Conformity	
Name:	Automated cell thawing device
Model:	
Date:	
Test Result:	Qualified
Inspection date:	

Warranty Card:

Warranty:

1.Enjoy ONE year free warranty for the whole machine from the date of purchase;

2.The following situations are not covered by the warranty:

A: Failure and damage caused by not operating correctly according to the manual;

B: Failures and damages caused by self-disassembly and repair;

C: Failures and damages caused by human factors;

D: If the warranty card cannot be presented or the instrument does not have the official sales seal.

Warranty				
No.	Fault cause	Repair dare	Repair man	Remark
1				
2				
3				
4				
5				
6				
7				



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TECHNOLOGY

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